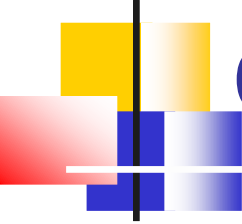


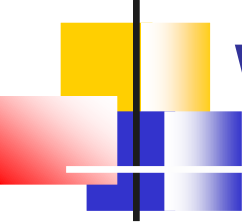
Ch1 Linux System Basics

Enyi Tang
Software Institute, Nanjing
University



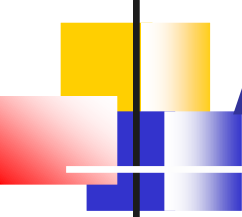
Content

- What is Linux
- Linux Installation
- Using the System
- Linux Programming Prerequisite
- Linux/UNIX Overview



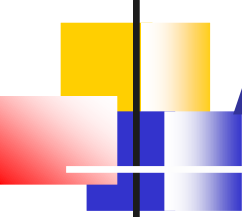
What is Linux?

- A free Unix-type operating system developed under the GNU General Public License.
 - Open source
 - Popular
 - Support most of the platforms available



A Short History of UNIX

- Multics: AT&T Bell Lab, GE, MIT
- UNIX: 1969, Ken Thompson, Dennis Ritchie
- Rewrite UNIX with C: 1973
- BSD: 1978, Berkeley Software Distribution
- System V: 1983
- Minix: 1987, Andrew Tannenbaum
- Commercial products
 - SunOS, Solaris, HP-UX, AIX, SCO UNIX
- Standards
 - SVID, IEEE POSIX, X/Open XPG4.2



A Short History of Linux(1)

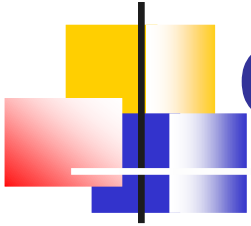
- 1984: Richard Stallman starts GNU project
 - GNU's Not Unix
 - <http://www.gnu.org>
- Purpose: Free UNIX
 - "Free as in Free Speech, not Free Beer"
- First step: re-implementation of UNIX Utilities
 - C compiler, C library
 - emacs
 - bash
- To fund the GNU project, the Free Software Foundation is founded
 - <http://www.fsf.org>



A Short History of Linux(2)

- 1991: Linus Torvalds writes 1st version of Linux kernel
 - Initially a research project about the 386 protected mode
 - Linus' UNIX -> Linux
 - Combined with the GNU and other tools forms a complete UNIX system
- 1992: First distributions emerge
 - Linux kernel
 - GNU and other tools
 - Installation procedure
- The rest is history...

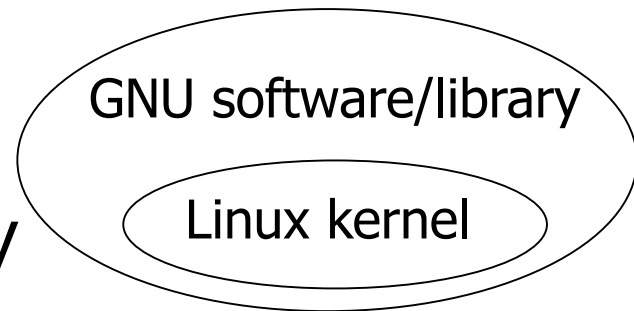




GNU & Linux

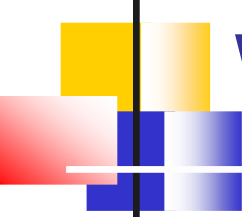
- GNU/Linux System

- Linux kernel
- GNU software/library



- Distributions:

- Ubuntu, Debian, Mint, Red Hat, Fedora, SuSe, Mandrake, Redflag...



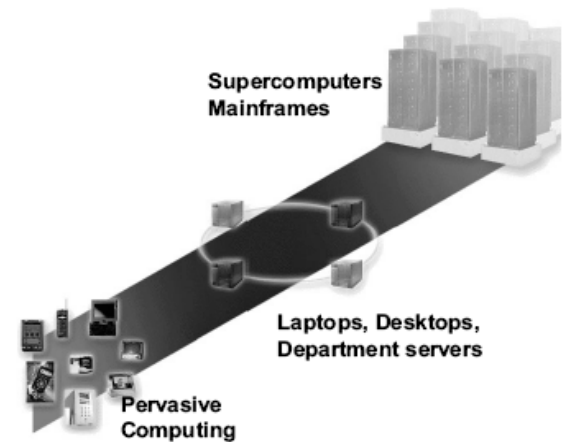
What is So Special for Linux?

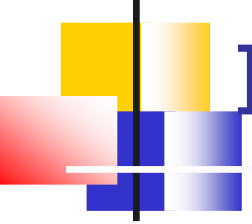
- Most software (including the Linux kernel) is **GPL**'ed (GNU General Public License)
 - <http://www.gnu.org/copyleft/gpl.html>
- Is called "copyleft" (instead of "copyright")
 - You may copy the software
 - You get the source code
 - You may alter the source code and recompile it
 - You may distribute the altered source and binaries
 - You may charge money for all this
- You only may not change the license
 - So all your customers have the same rights as you
 - So you really cannot make money from selling the software alone
- Other Open Source licenses (e.g. BSD) are also used



Linux Today

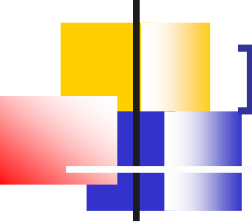
- Linux covers the whole spectrum of computing
 - Embedded devices
 - Laptops
 - Desktop systems
 - Development systems
 - Small and large servers
 - Megaclusters/supercomputers
- Linux is used throughout the world
 - ... and in space
- Linux is used by home users
 - ... and by some of the largest companies in the world
 - IBM
 - Boeing
 - NASA





Installation Methods

- Distributions:
 - Redhat -> Fedora
 - Debian
 - SuSe
 - Mandrake
 - Ubuntu
 -
- Live CD
- Using virtual machine
 - VMware, Virtual Box, etc.

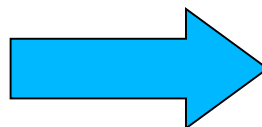


Installation Methods

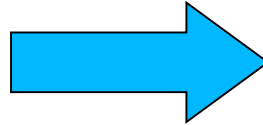
- Distributions:
 - Redhat -> Fedora
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 - SuSe
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 -
- Live CD
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Iceweasel 冰鼬



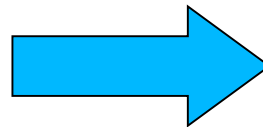
IceDove



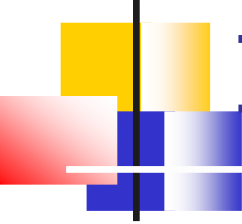
Iceape



Sunbird

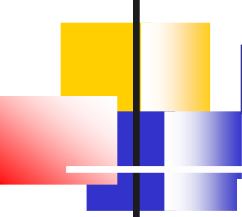


IceOwl



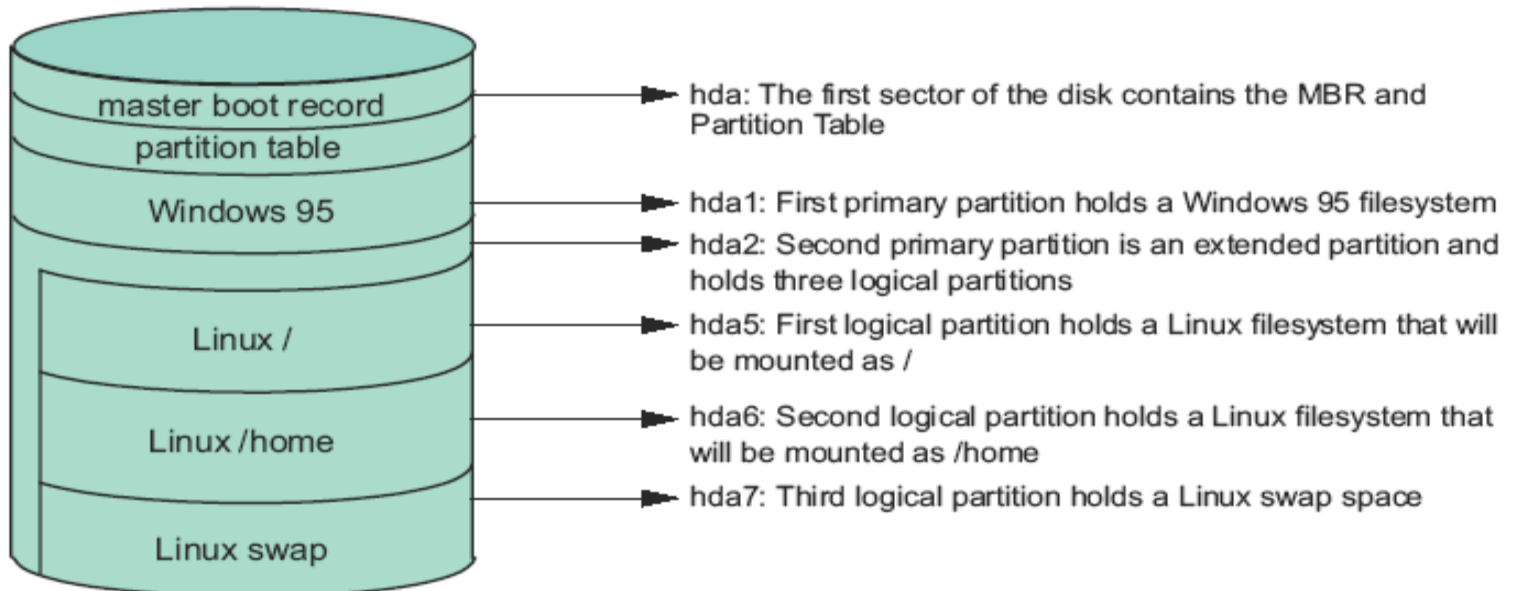
Installing Linux

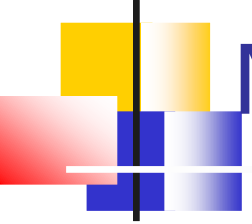
- Boot system from bootable media
- All installation programs need to perform essentially the same steps:
 - Choose language, keyboard type, mouse type
 - Create partitions **
 - Setup a boot loader **
 - Configure network
 - Configure user and authentication
 - Select package groups
 - Configure X
 - Install packages
 - Create boot disk



Partitioning Theory

- Partitioning is necessary on Intel-based computers
- Maximum of four primary partitions
- One primary partition may be an extended partition
- An extended partition can hold an unlimited amount of logical partitions (Linux: max 59)

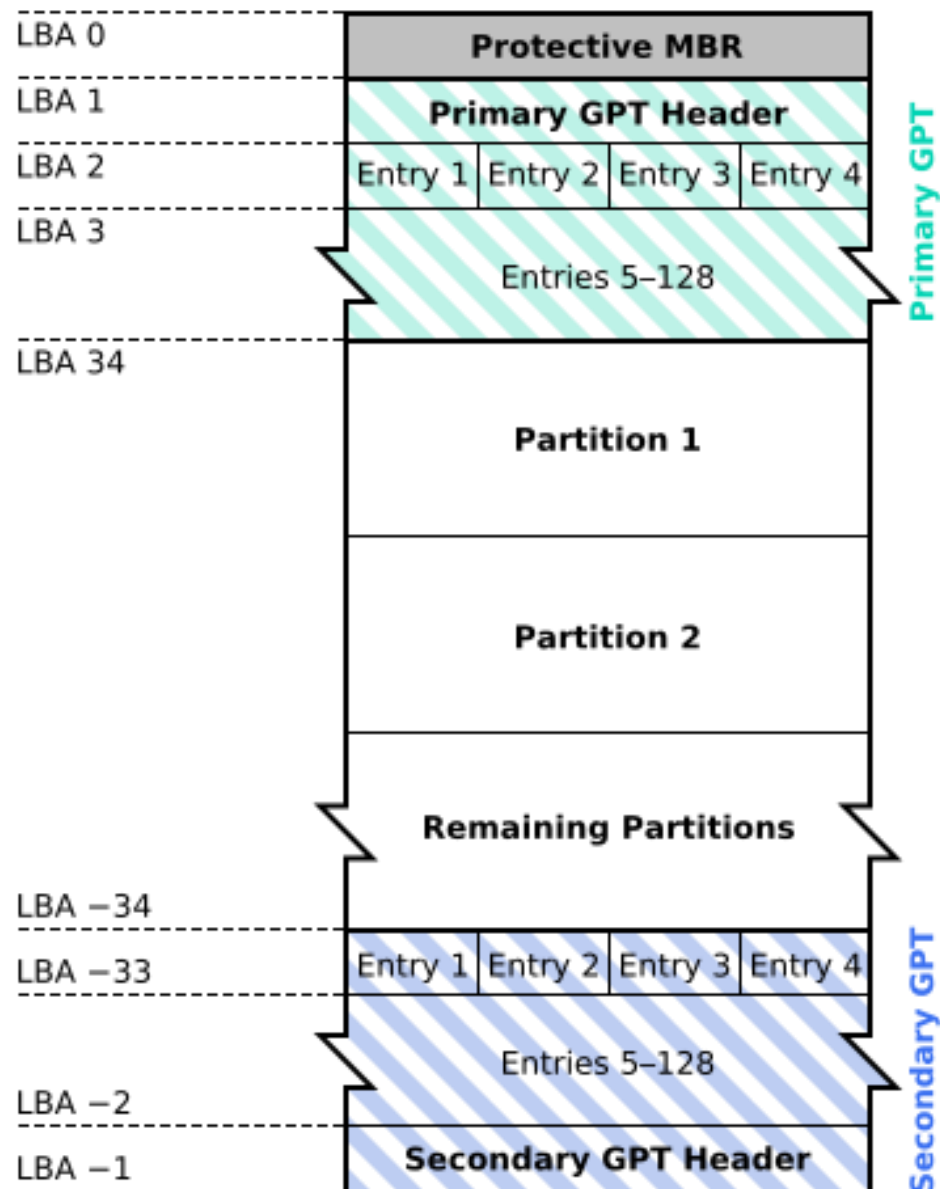


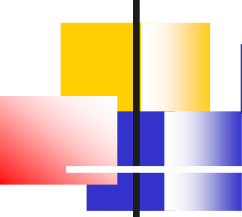


Master Boot Record

- Size: 512 bytes (first sector of hd)
- Addressed by BIOS
- Content:
 - 446 bytes program code (to boot an OS)
 - 64 bytes partition table with max. 4 entries
 - 2 bytes "magic number" (0x55AA)

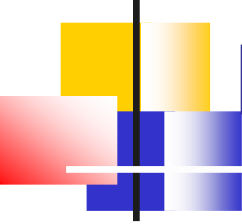
GUID Partition Table Scheme





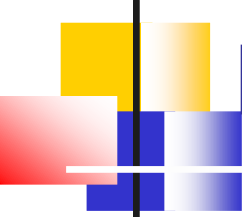
File System

- What is File System
 - 操作系统中负责存取和管理文件的部分
 - A collection of files and certain of their attributes. It provides a name space for file serial numbers referring to those files.
(susv3)
- File System in Linux:
 - VFS
 - EXT2, EXT3, FAT32, ...

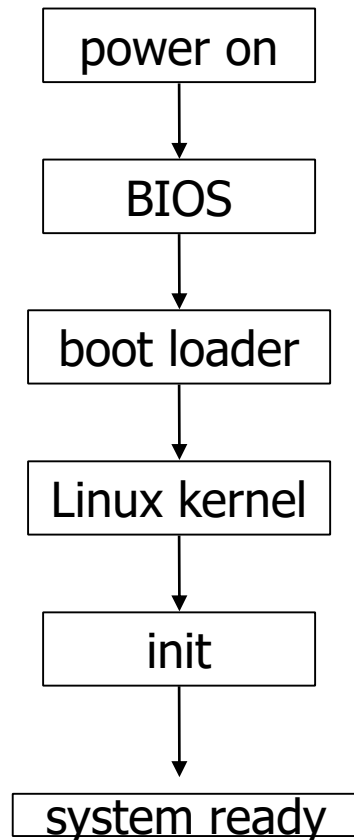


Disk Partitioning

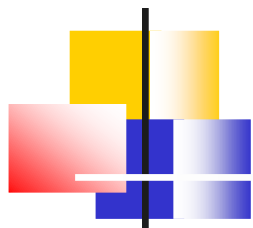
- At a minimum, create
 - `/`, 750MB (1.5G or more recommended)
 - Swap, size equal to amount of memory
- Recommended: `/boot` (16MB)
- May need/want to create other partitions:
 - `/usr`, `/usr/local`, `/var`, `/tmp`, `/opt`, `/home`
- Default partitioning program under Linux is `fdisk`
 - Distributions may add their own partitioning programs



Linux Startup Flow



- BIOS
 - Checks memory, loads options from non-volatile memory, checks for boot devices, loads MBR of boot device and executes it
- MBR
 - Contains a “boot loader” and the partition table
 - Traditionally set up by LILO/GRUB
- Boot loader
 - Loads the compressed kernel image into memory
 - The kernel uncompress itself and starts...
- Init process
 - Configuration file /etc/inittab
 - run levels



运算器
+
控制器

CPU

存储器

内存

输出设备

显示器

显卡

音箱/耳机

声卡

BIOS

主板

电源线

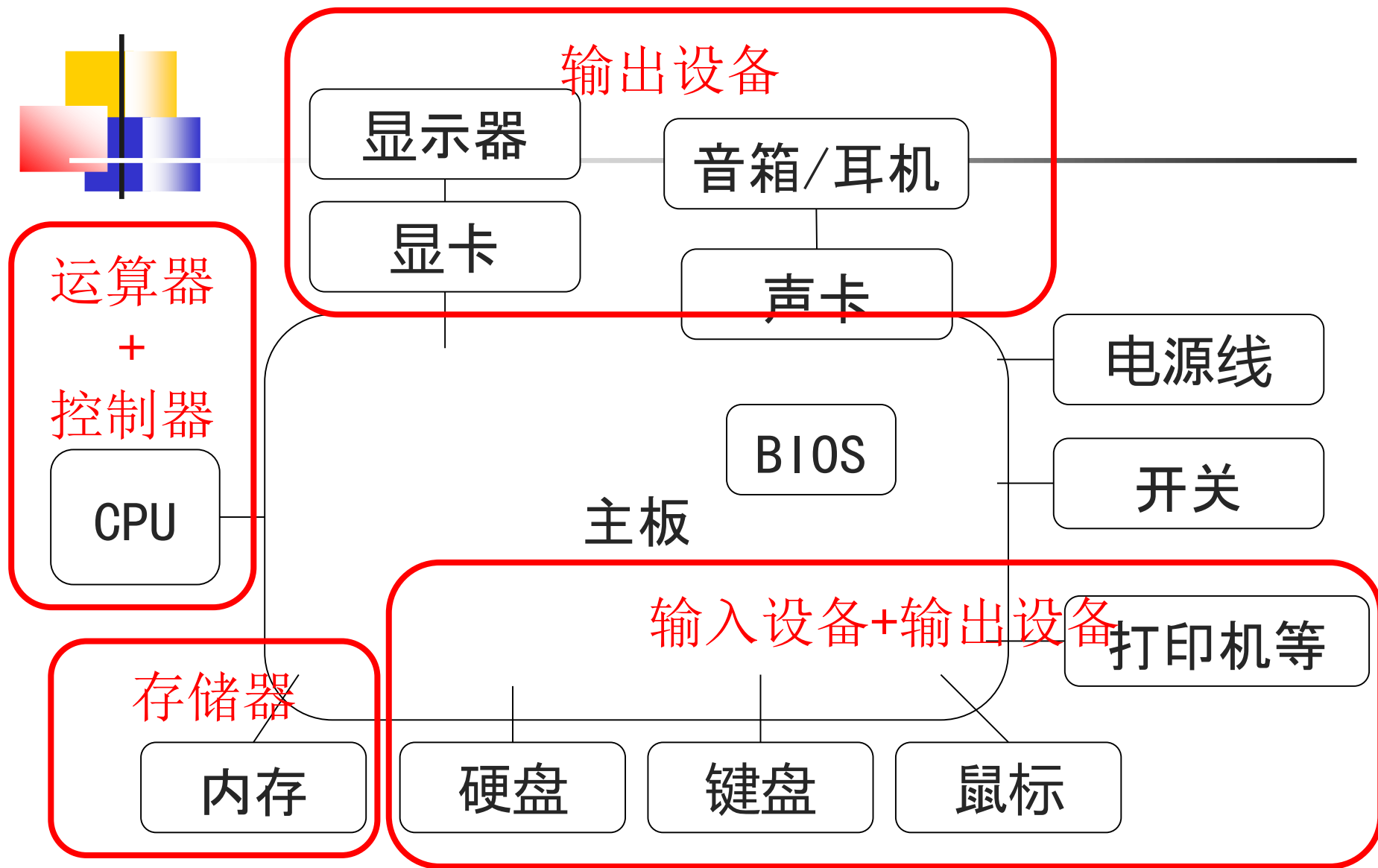
开关

输入设备+输出设备
打印机等

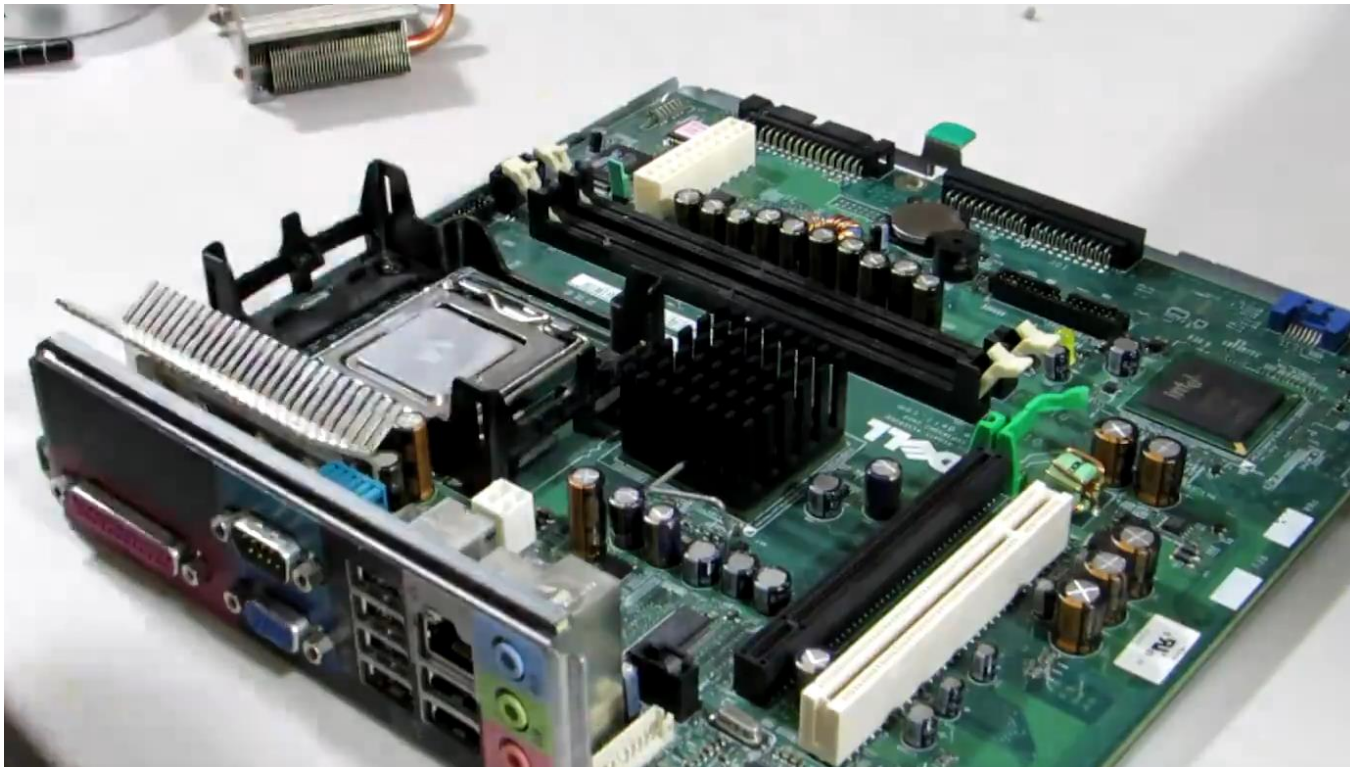
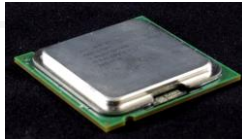
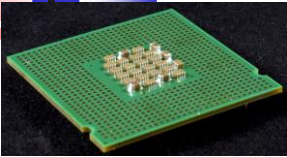
硬盘

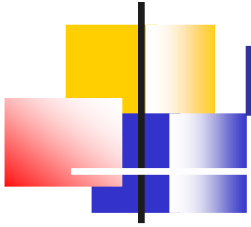
键盘

鼠标









Basic Input Output System (BIOS)

- Checks memory and hardware (POST)
- Loads options from non-volatile memory
 - Memory timings
 - Order of boot devices
- Checks for boot devices
 - Floppy disks, CD-ROM, Hard disks, etc.
- Load **MBR** of boot device and executes it

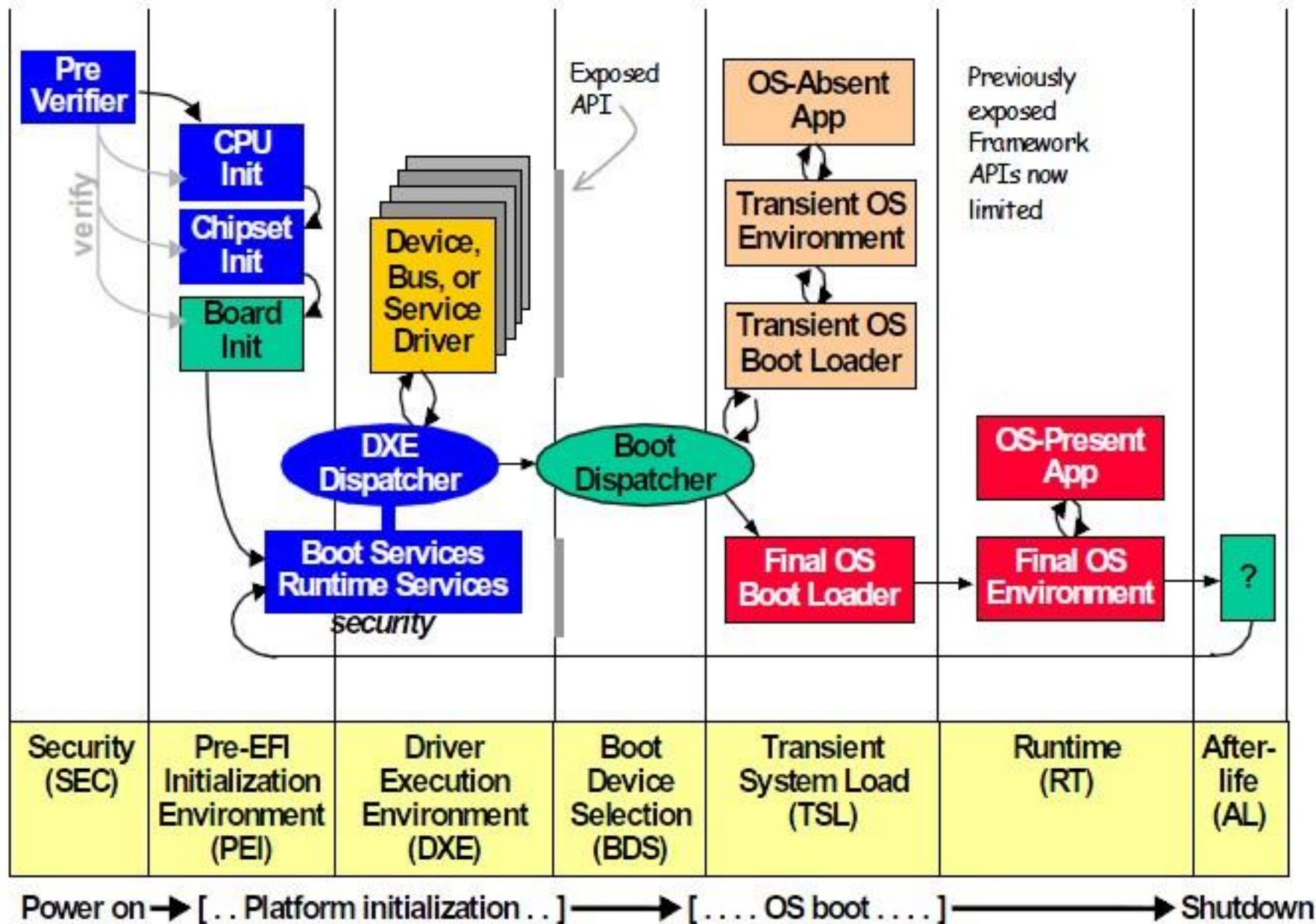
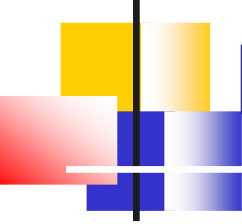
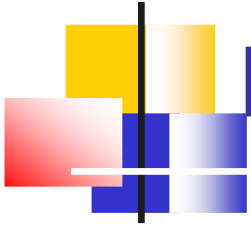


Figure 1-2. Framework Firmware Phases



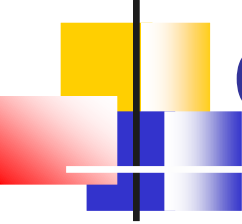
Boot loader

- A boot loader loads and starts the Linux kernel
 - Can pass boot parameters to the Linux kernel, such as device information
 - Can optionally load an Initial Root Disk
 - Can boot other operating systems as well
- Common Boot loaders:
 - LILO: Linux Loader
 - GRUB: Grand Unified Boot Loader
- Generally configured in `/dev/hda`, unless other boot loader is used.



LILO - Linux Loader

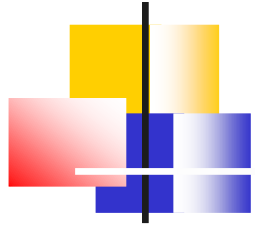
- LILO
 - A Program that configures the MBR according to the configuration file.
 - Must be run as *root* with the *lilo* command.
- *lilo* command Syntax:
 - `lilo [-v] [-v] [-C config-file] [-t]`
 - Configuration file: `/etc/lilo.conf`



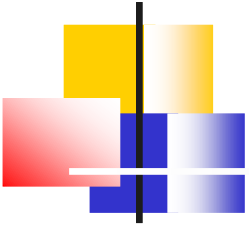
GRUB – GRand Unified Bootloader

- GRUB

- Program stored in MBR (first stage) and in /boot/grub (1.5th and second stage)
- Understand file system structure; no need to activate a configuration as with LILO
- Configuration file /boot/grub/grub.conf
- Installed in MBR with grub-install



-
- title Ubuntu, kernel 2.6.20-16-generic
 - root (hd0,1)
 - kernel /boot/vmlinuz-2.6.20-16-generic
root=UUID=3f784cd9-516f-4808-a601-
b19356f6bdea ro quiet splash locale=zh_CN
vga=0x318
 - initrd /boot/initrd.img-2.6.20-16-generic

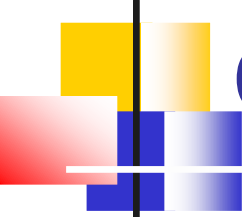


-
- title Microsoft Windows XP
Professional
root (hd0,0)
savedefault
makeactive
chainloader +1



Using the System

- Basic Knowledge
- Working with Files and Directories
- Working with Processes
- Linux Documentation

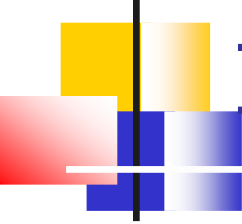


CLI vs GUI

```
brushington@MacBP:~$
```

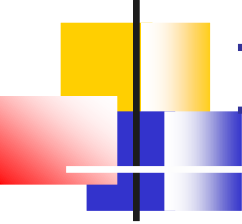
TABLE 2: Static Details

Component	Loop	LoopBound	Rec
adpcm	18	2424	
bs	1	4	
bsort	3	100	
cnt	4	10	
compress	7	50	
cover	3	120	



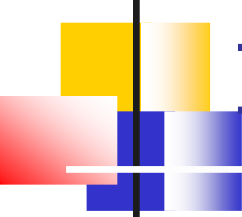
Installing Software on Linux

- From a tarball
 - `tar zxvf application.tar.gz`
 - `cd application`
 - `./configure`
 - `make`
 - `su -`
 - `make install`



Installing Software on Linux

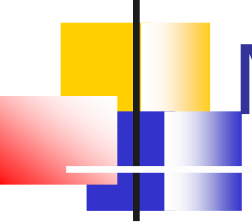
- From a tarball
 - `tar zxvf application.tar.gz`
 - `cd application`
 - `mkdir build`
 - `cd build`
 - `cmake ..`
 - `make VERBOSE=1`
 - `su -`
 - `make install`



Installing Software on Linux (cont'd)

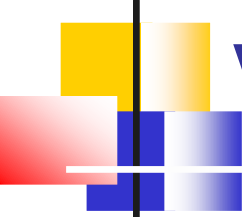
- apt-get command *
- dpkg
- aptitude

- yum + rpm
- RPM: RPM Package Management
 - rpm -q -a
 - rpm -ivh package-name
 - rpm -e package-name



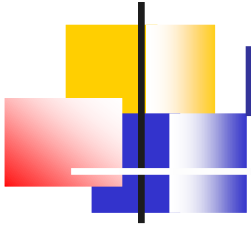
Multi-user and Multi-tasking

- Linux is a multi-user, multi-tasking operating system
 - Multiple users can run multiple tasks simultaneously, independent of each other.
- Always need to “log in” before using the system
 - Identify yourself with user name, password
- Multiple ways to log in to the system
 - Console: Directly attached keyboard, mouse, monitor
 - Serial terminal
 - Network connection



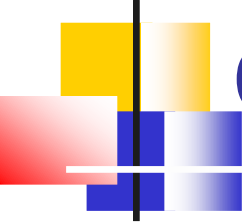
Virtual Terminal

- In most Linux distributions, the console emulates a number of virtual terminals
- Each virtual terminal can be seen as a separate, directly attached console
 - Different users can use different virtual terminals
- Typical setup:
 - VT 1-6: text mode logins
 - VT 7: graphical mode login prompt (if enabled)
- Switch between VTs with Alt-Fn (or Ctrl-Alt-Fn if in X)



Linux Commands

- Everything on a Linux system can be done by typing commands
 - the GUI (X-Window) is not needed for running a Linux System
- In order to type commands in X-Window you need to start a terminal emulator
- Command Prompt
 - Can be configured yourself
 - \$ - "logged in as a regular user",
 - # - "logged in as root"



Command Syntax

- Linux commands have the following format:

\$ command option(s) argument(s)

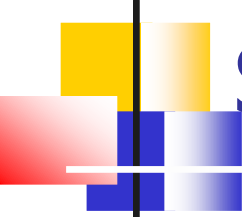
- Examples:

\$ ls

\$ ls -l

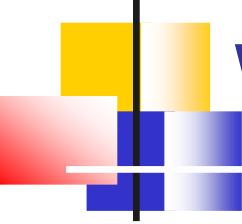
\$ ls /dev

\$ ls -l /dev



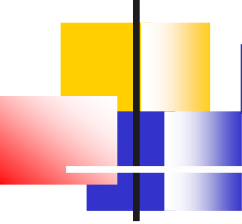
Some Basic Linux Commands

- passwd: Change your password
- mkpasswd: Generate a random password
- date, cal: Find out today's date and display a calendar
- who, finger: Find out who else is active on the system
- clear: Clear the screen
- echo: Write a message to your screen
- write, wall, talk; mesg
-



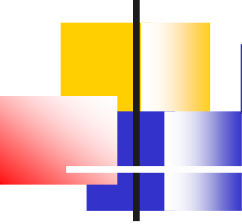
Working with Files & Directories

- What is a file?
 - A collection of data;
 - An object that can be written to, or read from, or both. A file has certain attributes, including access permissions and type. (susv3)
- File structure
 - Generally: byte stream, record sequence, record tree
 - In Linux: byte stream



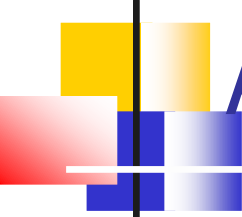
File Types

- regular file
 - Text or code data; no particular internal structure
- character special file
- block special file
 - Special files: represent hardware or logical devices
 - Found in directory called /dev
- socket
- symbolic link
- Directory
 - A table of contents; a list of files in that directory

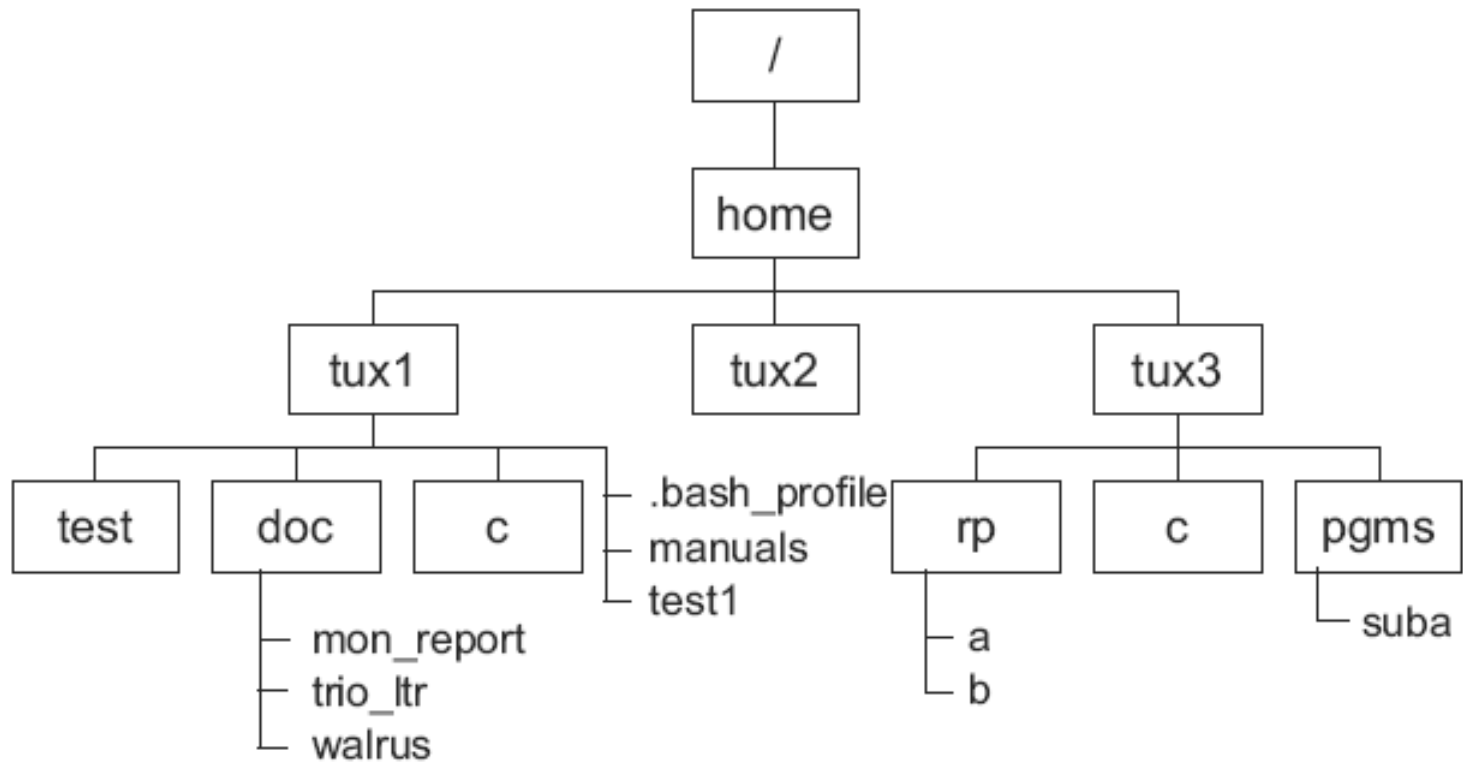


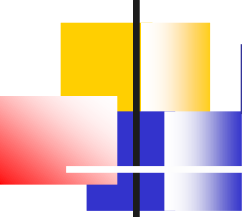
Directory Structure

- All Linux directories are contained in one, virtual, “unified file system”.
- Physical devices are mounted on mount points
 - Floppy disks
 - Hard disk partition
 - CD-ROM drives
- No drive letter like A:, C:, ...

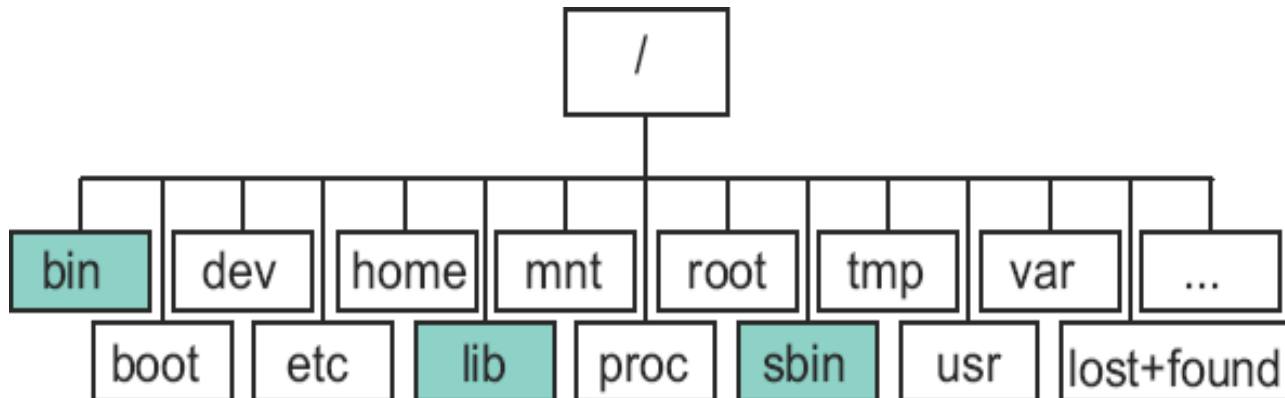


An Example of Directory Structure

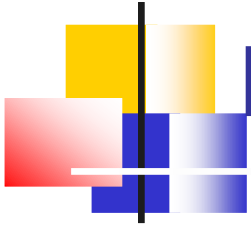




Main Directories in Linux

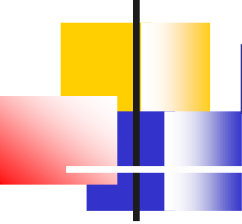


Linux Filesystem Hierarchy Standard:
<http://www.pathname.com/fhs>



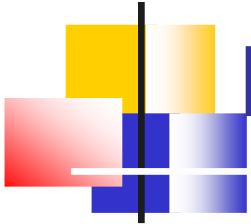
Basic Commands(1)

- Commands used with directories:
 - pwd: print working directory
 - cd: change directory
 - mkdir: make directory
 - rmdir: remove directory
 - ls: list the contents of directories
 - -l, -a, -R options



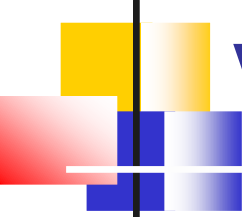
Basic Commands(2)

- commands used with files:
 - touch: update the access and/or modification time of a files
 - cp: copy files
 - mv: move and rename files
 - ln: link files
 - rm: remove files
 - cat: print file contents
 - more/less: display files page by page



File Permission

- File Permissions help you protect your files against other users on the system
- Three access levels:
 - User: The user that created the file
 - Group: All users in the group that owns the file
 - Others: All others
- Three permissions:
 - Read (**r**): Read content of file or list content of directory
 - Write (**w**): Change content of file or create/delete files in directory
 - Execute (**x**): Execute file as program or use directory as active directory



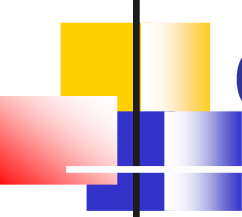
Viewing File Permissions

To show the permissions of a file, use the **ls** command with the **-l** option

```
$ ls -l
```

-rw-r--r--	1	tux1	penguins	101	Jun 5 10:03	file1
-rw-r--r--	1	tux2	penguins	171	Jun 4 10:23	file2
drwxr-xr-x	2	tux1	penguins	512	Jun 7 11:13	mydir

File type		owner	group	size	mtime (modification time)	name
	permissions	link counter				



Changing Permissions

- The change mode command:

```
$ chmod <who operator what> filename
```

who:

u = owner of file

g = group

o = other users on the system

a = all (owner+group+others)

operator:

+ = add permission

- = remove permission

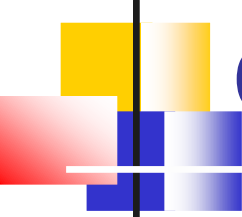
= = clear permissions and set to mode specified

what:

r = read

w = write

x = execute



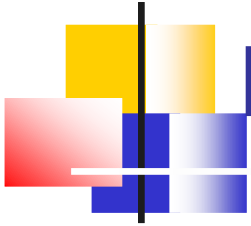
Changing Permissions (cont'd)

- File and directory permissions can also be specified as an octal number:

	User	Group	Other
Symbolic notation	rwX	rw-	r-x
Binary	111	110	101
	4+2+1	4+2+0	4+0+1
Octal	7	6	5

```
$ chmod 765 file
```



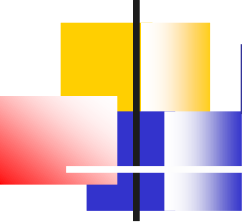


Default File Permissions

- The default permission for newly created files and directories are:

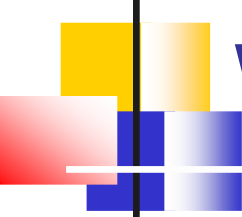
File: -rw-r--r-- 644

Directory: drwxr-xr-x 755



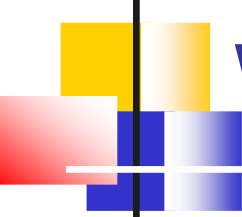
Editing Files

- vi
- emacs
- gedit
- ...



Working with Processes

- What is a process?
 - A process is a task.
 - 进程是一个正在执行的程序实例。由执行程序、它的当前值、状态信息以及通过操作系统管理此进程执行情况的资源组成。
 - An address space with one or more threads executing within that address space, and the required system resources for those threads. (susv3)

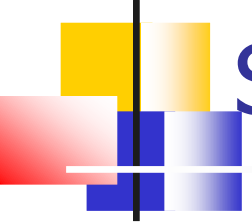


Working with Processes (cont'd)

- A running program is an example of a process

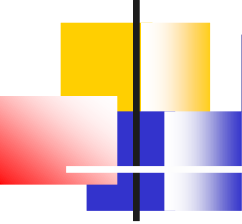
The Process Environment	
Program name	User and group ID
Data	Process ID (PID)
Open Files	Parent PID (PPID)
Current Directory	Program variables

- A shell is a process that reads your commands and start the appropriate process.
 - `echo $$`



Starting and Stopping a Process

- All processes are started by other processes
 - Parent/Child relationship
 - One exception: init (PID 1) is started by the kernel itself
 - A tree hierarchy
- A process can be terminated because of two reasons:
 - The process terminates itself when done.
 - The process is terminated by a signal from another process

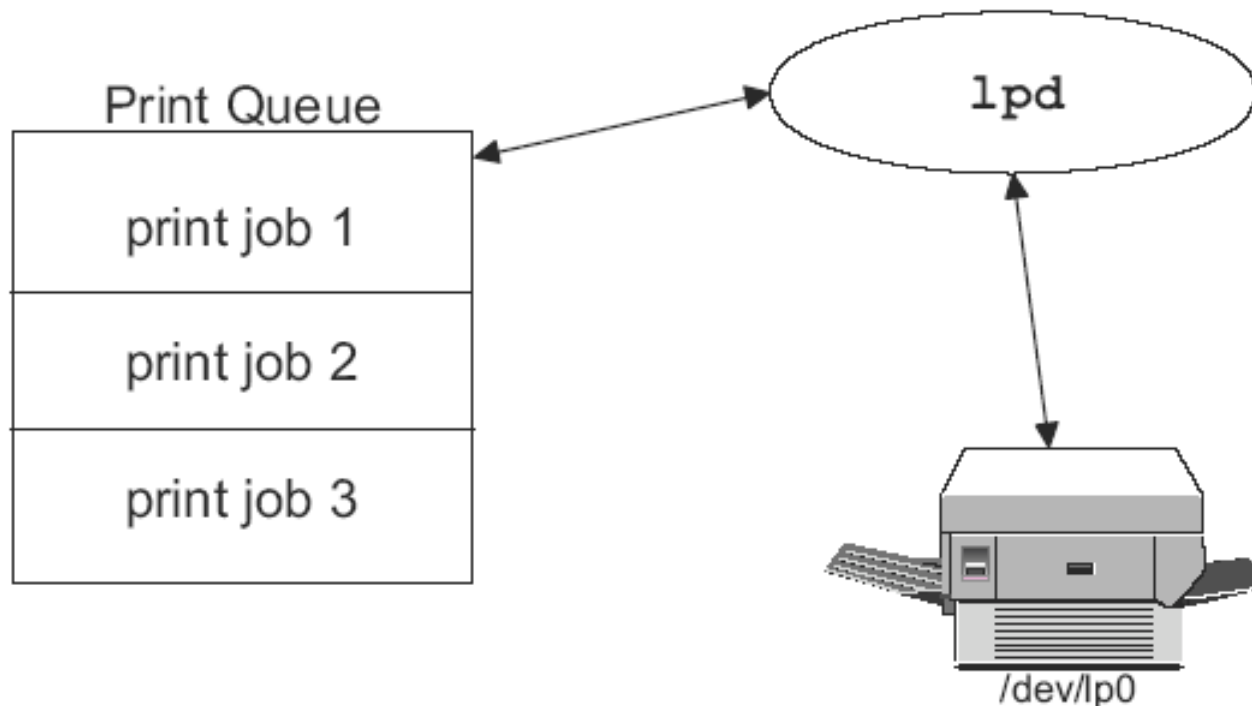


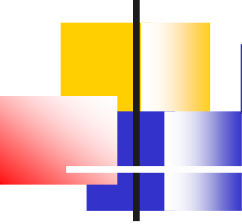
Basic Commands

- ps: report process status
- pstree: display a tree of processes
- jobs, fg, bg, <ctrl-z>: job controlling
- kill:
- nohup: run a command, ignoring hangup signals
- nice, renice:
- top: display top CPU processes

Daemons

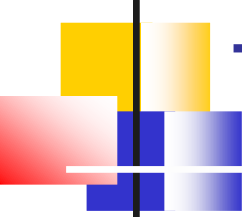
- The word "Daemon" refers to a never-ending process, usually a system process that controls a system resource such as the printer queue or performs a network service





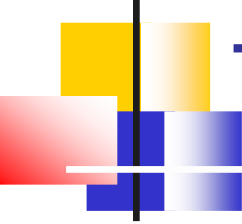
How to Find Help?

- “man” command
- “info”
- command --help
- HOWTO Documentation
- Refer to Internet



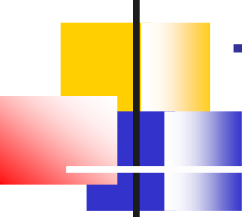
The man command

- With the man command you can read the manual page of commands
- Manual pages are stored in `/usr/man`
- The manual page consists of:
 - Name: The name of the command and a online description
 - Synopsis: The syntax of the command
 - Description: Explanation of how the command works and what it does
 - Files: The files used by the command
 - Bugs: Known bugs and errors
 - See also: Other commands related to this one



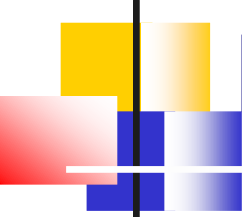
The man command (cont'd)

- The “-k” option
 - `man -k print`
- Manual pages are divided in 8 sections:
 1. User commands
 2. System calls
 3. Libc calls
 4. Devices
 5. File formats and protocols
 6. Games
 7. Conventions, macro packages and so forth
 8. System administration
- To select correct section, add section number:
 - `man 1 passwd`, `man 5 passwd`



The info command

- A program for reading documentation, sometimes a replacement for manual pages
- Information for info is stored in /usr/info
- Some info commands:
 - space: next screen of text
 - delete: previous screen of text
 - n: next node
 - p: previous node
 - u: up node
 - q: quit info
 - <tab>: skip to next menu item

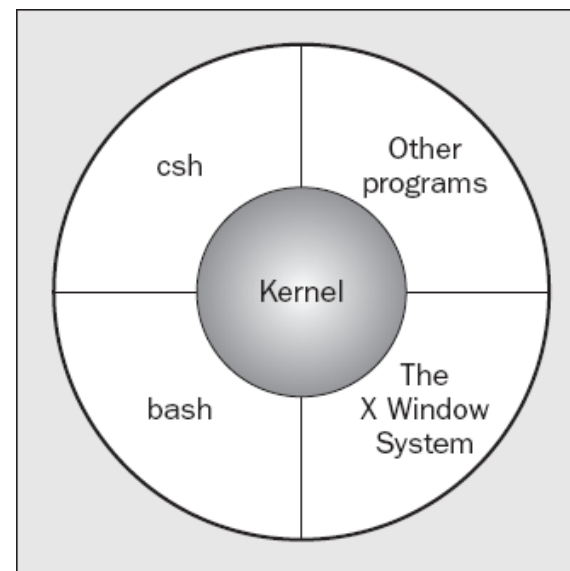


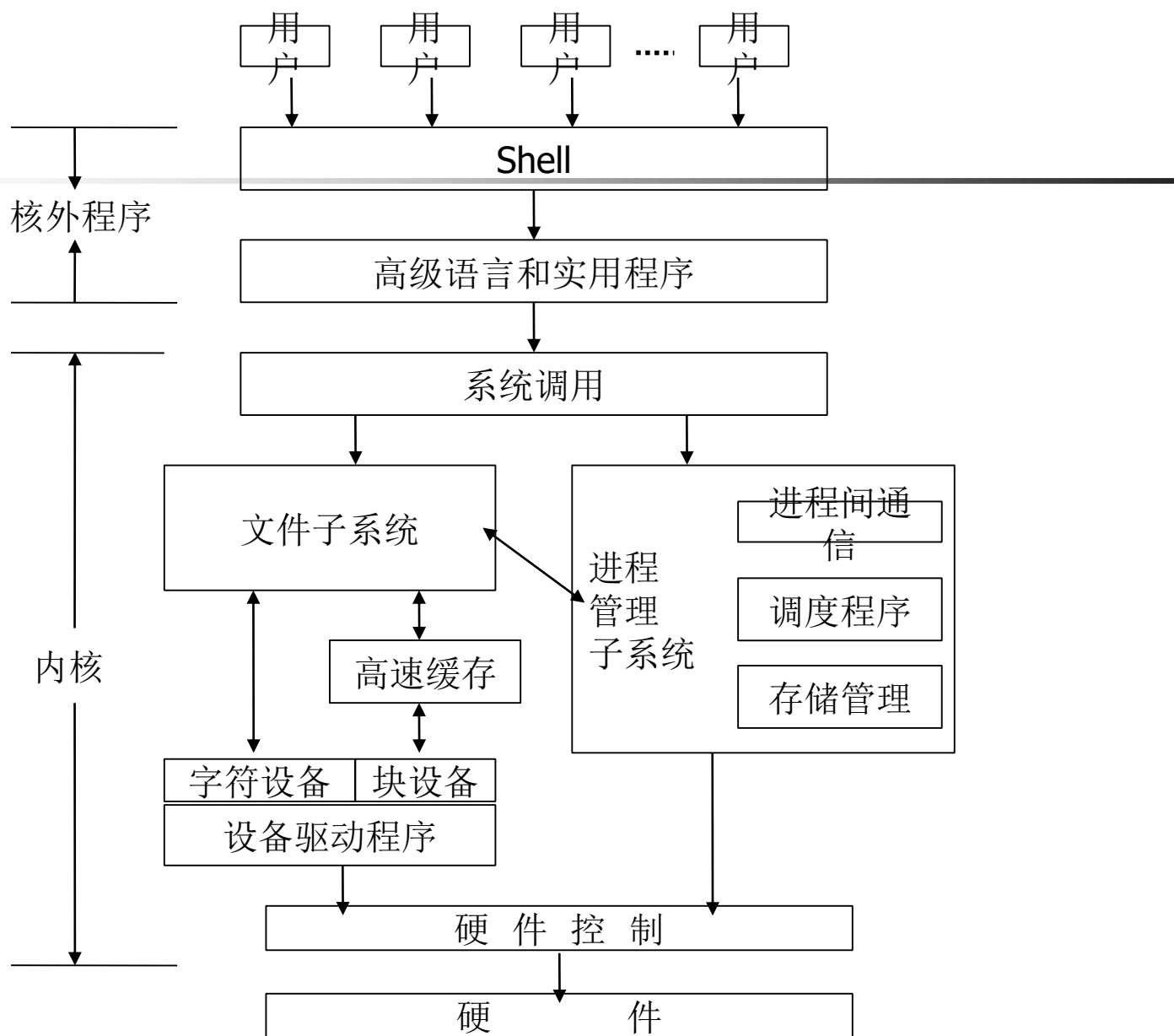
Exercises

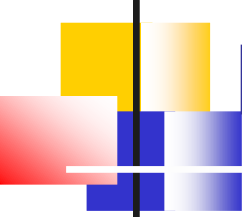
- 浏览网站：
 - <http://www.gnu.org>
 - <http://www.linux.org>
- 安装一种Linux Distribution，然后在其上安装一些需要的软件
- 学习Linux基本命令的使用
- 复习C程序设计语言

UNIX Overview

- 早期的UNIX
 - 一个简单的文件系统
 - 一个进程子系统和一个Shell
- 内核和核外程序







Programmer's Viewpoint

